Validity-Assured Regeneration of Parametric Solid Modeling

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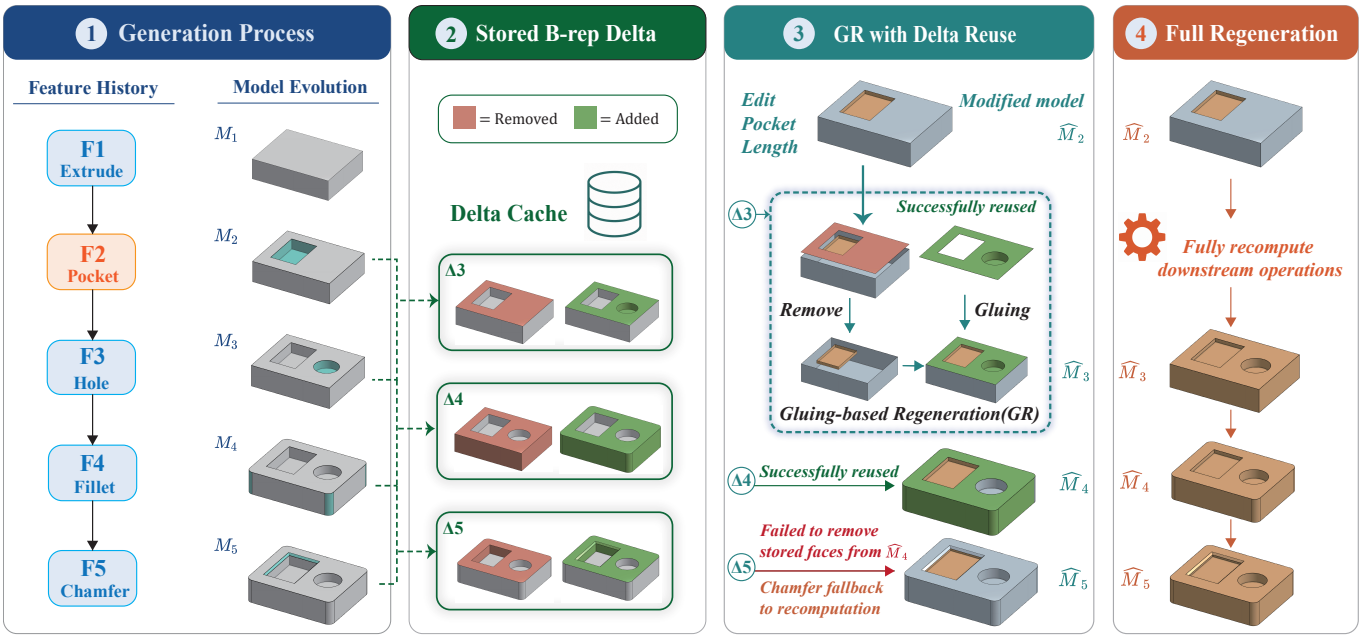

Fig. 1. During the modeling process, feature parameters are often adjusted to satisfy different design requirements. Full regeneration then triggers a sequential re-execution of the history from the modified step onward, which can be computationally expensive. Our method aims to accelerate this process by recording B-rep deltas and performing gluing-based regeneration, while guaranteeing consistency with the result of full regeneration.

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Feature-based parametric modeling remains the primary approach for product modeling. It represents model construction as a sequence of parameterized features and re-executes the sequence whenever the parameters are modified, a procedure known as regeneration. Regeneration is triggered frequently during model editing, making its efficiency crucial in product design. Despite its practical significance, regeneration has received limited attention in the academic literature. Moreover, existing criteria for determining reusable previously computed data for regeneration may lead to incorrect results, making validity crucial. Yet validity is rarely discussed or theoretically justified in prior regeneration methods. In this paper, we present a validity-assured method that accelerates regeneration. Specifically, we (i) propose two novel locality criteria under which information from generation can be reused to accelerate regeneration while guaranteeing correctness, and (ii) provide a rigorous theoretical proof of this correctness guarantee. The validation dataset and associated tools for this paper are available at https://mechano-pesudo.github.io/VAR.

CCS Concepts: • **Applied computing** → **Computer-aided design**; • **Theory of computation** → **Computational geometry**.

Additional Key Words and Phrases: Parametric solid modeling, Regeneration, Validity Assurance

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1 Introduction

Feature-based parametric solid modeling has become the de facto paradigm for model construction in mainstream CAD/CAM systems, including CATIA, NX, SolidWorks, and Creo [Camba et al. 2016; Dassault Systèmes SolidWorks Corporation 2020; PTC 2025; Siemens PLM Software 2007]. In such systems, a model is created by sequentially applying parameterized operations, which are recorded in a history (or feature) tree and connected through explicit parent–child dependencies [Bidarra et al. 2005; Camba et al. 2016; Yang et al. 2012].

In parametric CAD design, model updates are commonly achieved by modifying feature parameters [Allen 2007], as shown by the examples in Fig. 1. These parameter changes often propagate through the history tree and may trigger a large number of Boolean operations. Recomputing all Boolean operations from scratch—i.e., performing a full Boolean regeneration (FBR)—would incur prohibitive computational costs and severely compromise system interactivity in industrial settings [Allen 2007; Wang et al. 2015; Zou 2025]. Consequently, commercial CAD systems adopt a variety of regeneration methods that *reuse* previously computed results and restrict recomputation to affected regions of the model [Rameau 2017; Rameau et al. 2015; Ran et al. 2012]. Actually, modern CAD systems perform local regeneration rather than FBR. To support interactive performance, *locality criteria* are employed to determine which Boolean operations and geometric entities must be recomputed and which previously computed results can be reused. While these criteria significantly reduce computational cost, however, existing locality criteria may still lead to regenerated models that are inconsistent with the results of FBR, thereby compromising model validity. Despite its practical importance, the problem remains largely unresolved: existing solutions are only vaguely described in patents, and the academic literature has paid limited attention to the systematic design of locality criteria that can *guarantee* validity with respect to full Boolean regeneration results.

In this paper, we revisit this fundamental problem from a more principled perspective. Our goal is to design locality criteria such that local regeneration is provably equivalent to FBR. Through extensive analysis, we propose two novel locality criteria that provably ensure consistency between local regeneration and FBR. Furthermore, we develop a theoretical framework and provide rigorous proofs to establish the validity of the proposed approach. To the best of our knowledge, this work represents the first study to present a validity-assured regeneration method grounded in equivalence to full Boolean regeneration, and we believe it will inspire further research on robust regeneration mechanisms in parametric solid modeling systems.

2 Related Work

We first review prior work on regeneration acceleration, followed by the persistent naming problem in regeneration.

2.1 Regeneration Acceleration

Since most regeneration-acceleration methods lack public implementations and several are patent-based, we discuss them conceptually rather than as directly reproducible experimental baselines.

A classical approach accelerates regeneration by reorganizing the feature-evaluation structure. [Chin et al. 2007] reorganizes linear histories into feature chains based on feature scope, allowing regeneration order to differ from creation order, while [Rameau 2017] replaces ordered lists of basic volumes with balanced binary trees to achieve faster and more uniform declarative updates.

Another direction reduces recomputation directly. [Ran et al. 2012] proposes local regeneration by reconstructing only affected features, aiming to reduce update time while producing the same final model as a full rebuild; however, the patent description is too brief to support viewing it as a method with a theoretical correctness guarantee. [Yang et al. 2012] updates a feature's boundary contribution by storing intersection face portions and using remove/re-add for modification, thereby reducing dependence on full history replay. This comes at the cost of additional intersection bookkeeping and ongoing dependence on naming/matching, along with unresolved child-feature repositioning when references break. [Rameau et al. 2015] accelerates updates by encoding solids as cell graphs and reusing previous results via graph-rewriting rules when an edit is independent and passes a 3D-compatibility check, avoiding full replay.

2.2 Persistent Naming

In parametric modeling, downstream features often reference topological entities in earlier B-rep states for attachment or placement. When the parameters of upstream operations are modified, regeneration may cause these entities to change or disappear. The system must then identify corresponding entities in the updated model to preserve feature semantics and design intent. This challenge is known as the persistent naming problem.

Persistent naming has long been a fundamental issue in feature-based parametric modeling. Representative studies include generic naming [Capoyleas et al. 1996], the topological ID system [Kripac 1997], history-based face naming [Wu et al. 2001], and an intent-preserving lineage-based referencing DSL [Cascaval et al. 2023]; broader reviews are given in [Farjana and Han 2018; Marcheix and Pierra 2002]. In the following, we assume that each topological entity in a B-rep is assigned an unambiguous name and that downstream references can be updated automatically during regeneration.

3 Formulation

In this section, we briefly review the parametric modeling process, fix the notation used throughout the paper, and provide an overview of regeneration algorithm.

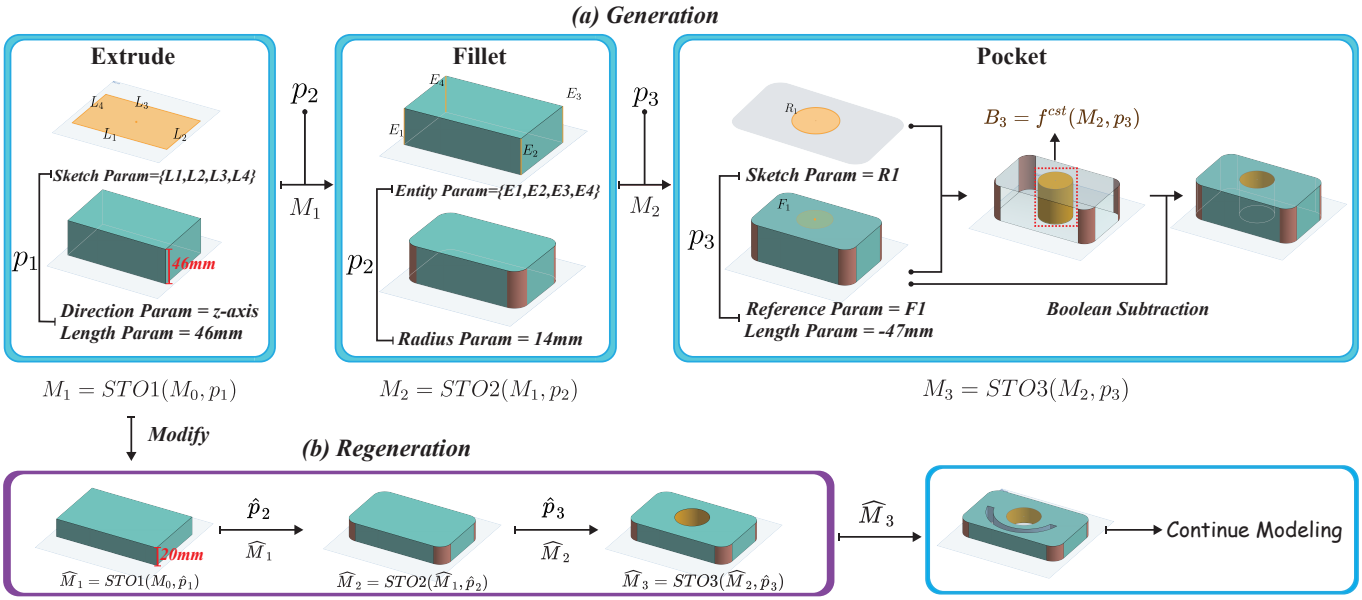


Fig. 2. (a) Generation process. In the pocket step, the sketch parameters, pocket length, and topological references jointly define the intermediate solid B_3, which is then subtracted from the input solid M_2 to produce the output. This procedure is encapsulated as a Type-I STO. The figure also illustrates the encapsulation of a Type-II STO, namely the process in which a fillet is applied to M_1. (b) Regeneration process. After the length parameter in the first modeling step is changed to 20 mm, subsequent operation parameters may be updated accordingly, and each affected step is re-executed with the updated parameters.

3.1 Generation

Generation refers to the construction of a CAD model through an ordered sequence of parametric modeling commands(Fig. 2(a)). This sequence represents the modeling history of the model.

Meta notation. In the CAD literature, the modeling history of a model M is commonly represented using the following meta notation [Fan et al. 2025; Wu et al. 2021]:

$$M = [f_1, f_2, \ldots, f_n],$$

where each f_i is a command/operation, such as Line, Arc, Extrude, Fillet, or Selection, which includes modeling parameters.

3.2 Solid-Transform Operation (STO)

Meta notation is not suitable for tracking how the solid evolves throughout the modeling process, since operations such as Line and Arc create sketches without directly modifying the 3D solid itself. We therefore focus on the feature operations that directly affect the solid and further classify them into two categories:

(1) Type-I operations first generate an intermediate solid and then combine it with the input solid through a Boolean operation, such as Extrude, Revolve, Sweep, Loft.
(2) Type-II operations do not generate intermediate shapes but instead act directly on entities of the input solid to produce the output, such as Fillet and Chamfer.

Our notation. We abstract the overall process by which a Type-I or Type-II operation acts on an input solid as a Solid-Transform Operation (STO). The input to an STO (S) is divided into two components: the input solid (M_{in}) and the STO parameters (p). After executing the STO, a new solid (M_{out}) is produced. We represent the process as below:

$$M_{out} = S(M_{in}, p).$$

For convenience, STOs induced by Type-I and Type-II operations are referred to as Type-I STOs and Type-II STOs, respectively.

To make the notation precise, we detail the complete procedure of a Type-I STO. It first constructs an intermediate solid B using the topological references in M_{in} and other operation parameters specified by p. We denote this construction process by f^{cst}:

$$B = f^{\text{cst}}(M_{in}, p).$$

The intermediate solid B is then combined with M_{in} through a Boolean operation to produce M_{out}, where the Boolean type is also specified by p. Fig. 2(a) illustrates the parameters and processes of different STOs.

In this work, the modeling of a part is viewed as the result of a sequence of STOs. We then represent the process of generation as below:

$$\begin{aligned} M_n &= S_n(M_{n-1}, p_n) \\ &= S_n(S_{n-1}(\ldots S_2(S_1(M_0, p_1), p_2)\ldots, p_{n-1}), p_n). \end{aligned}$$

Here, M_0 denotes the initial empty solid, and M_n the resulting solid after applying the first n STOs. S_n denotes the n^{th} STO, which transforms M_{n-1} into M_n, with parameter set p_n. For a Type-I STO, B_n denotes the intermediate solid generated at step n.

In this paper, we primarily focus on modeling sequences composed of Type-I STOs. Type-II STOs can also be handled within our framework, as discussed in Section 4 of the supplementary material.

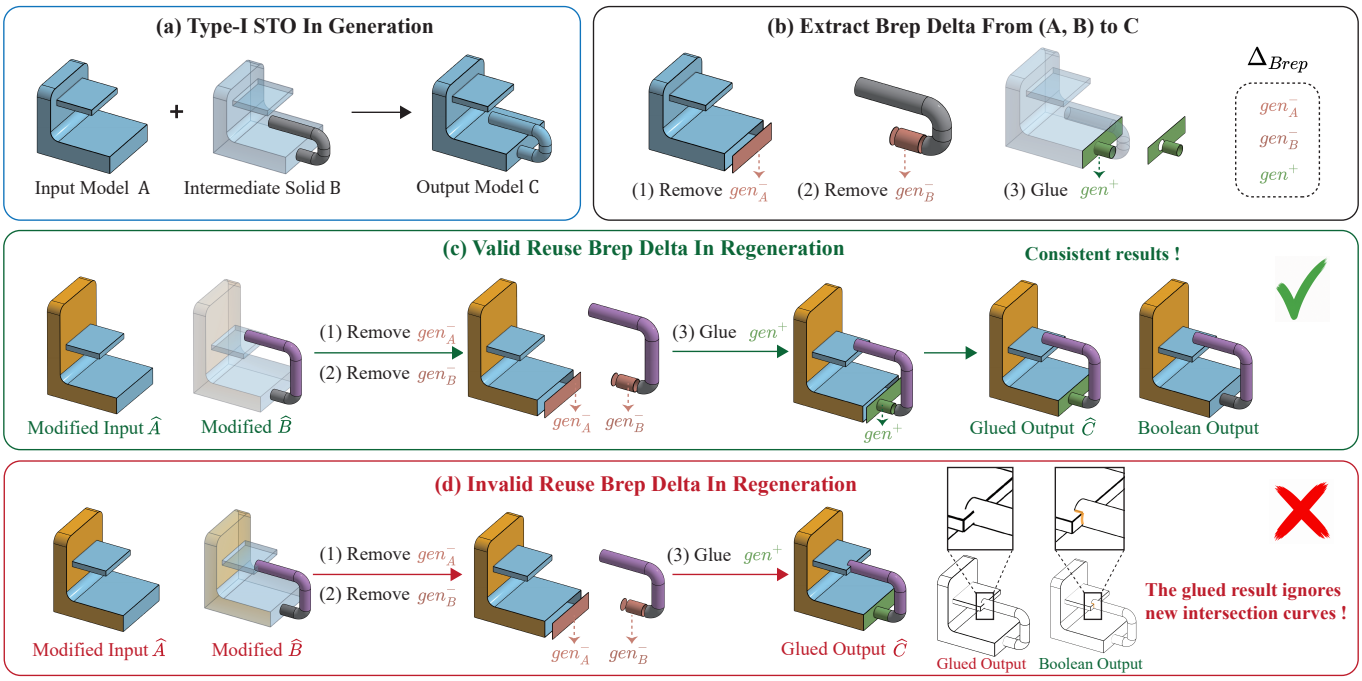


Fig. 3. (a) A Type-I STO involving Boolean union during generation. (b) The B-rep delta extracted from the Boolean process from (A, B) to C. (c) When the input changes from (A, B) to $(\widehat{A}, \widehat{B})$, reusing the recorded B-rep delta through B-rep removal and topological gluing yields the same result as Boolean computation. (d) In this case, directly reusing the recorded B-rep delta ignores newly created intersection curves, and thus produces a B-rep that is incorrect in both topology and geometry.

3.3 Full Boolean Regeneration (FBR)

Regeneration refers to the process in which the user modifies the modeling history of a model, triggering a sequential re-execution of the history to update the resulting model (Fig. 2(b)).

When a user edits an existing model with n STOs by modifying the parameters at step $k (k < n)$ from p_k to $\widehat{p}_k$, the system first rolls back from M_n to M_{k-1}, and then updates M_k to $\widehat{M}_k$ using the updated parameters:

$$\widehat{M}_k = S_k(M_{k-1}, \widehat{p}_k).$$

Due to the parent–child dependencies between features [Camba et al. 2016], modifying p_k may also trigger parameter updates in subsequent affected features, depending on the persistent-naming mechanism [Kripac 1997]. Therefore, for each step j, the model is regenerated by re-executing S_j with the updated parameters $\widehat{p}_j$:

$$\widehat{M}_j = S_j(\widehat{M}_{j-1}, \widehat{p}_j), \qquad j = k+1, \ldots, n.$$

This process may involve time-consuming Boolean computations in Type-I STOs. We refer to recomputing the Boolean operation of a single Type-I STO as *Boolean regeneration* (BR), and to applying BR to all downstream STOs after the modified feature as *full Boolean regeneration* (FBR).

3.4 Notation and Problem Formulation

Boundary Representation(B-rep). A B-rep represents a solid by explicitly describing its boundary topology and geometry. For a solid model M, we denote its B-rep by $Brep(M)$, its boundary by ∂M, and its boundary face set by F_M. We regard F_M as a face set associated with the adjacency inherited from $Brep(M)$, so that any subset of F_M retains the induced topological adjacency from the original solid.

Topological gluing operation. In this paper, we use *topological gluing operation* to denote the process of constructing a new B-rep from faces or shells according to prescribed topological connectivity. Under this view, a solid B-rep can be regarded as the result of gluing together its boundary faces, as illustrated in Fig. 4. We denote by $\mathrm{Glue}(\mathcal{F})$ the B-rep obtained by gluing a face set $\mathcal{F}$ with known topological connectivity.

Other Notations. In the following discussion, unless otherwise specified, we regard a solid M as the closed 3D region it occupies in $\mathbb{R}^3$, and denote by ∂M the boundary point set of this region. Thus, saying that ∂M is connected means that the solid has a single connected boundary component, rather than multiple disconnected boundary components, as illustrated in Fig. 5. For a topological face f, we use $|f|$ to denote the point set covered by f, including its boundary points. For a set of faces $\mathcal{F}$ we define $|\mathcal{F}| = \bigcup_{f \in \mathcal{F}} |f|$.

Face Correspondence in Boolean Operations. For a Boolean union $C = A \cup B$, a face $f \in F_A$ is considered unaffected if $|f| \cap B = \emptyset$, and similarly $g \in F_B$ is unaffected if $|g| \cap A = \emptyset$. We identify such unaffected faces with their corresponding counterparts in F_C, allowing set differences among F_C, F_A, and F_B to be well defined.

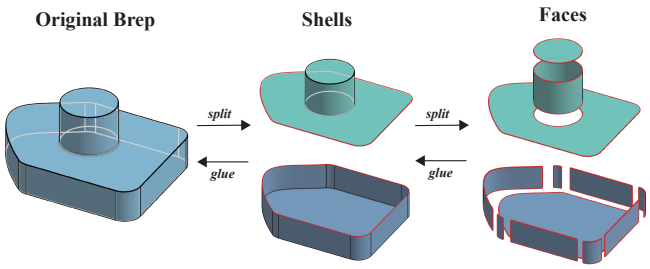


Fig. 4. A B-rep can be regarded as the result of a topological gluing operation applied to its constituent sub-shells, or equivalently, to its boundary faces.

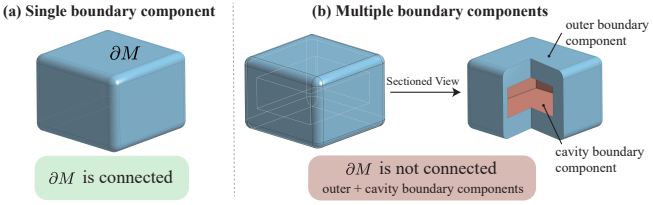


Fig. 5. Examples illustrating (a) a solid M with a connected boundary and (b) a solid M with a cavity, whose outer boundary and cavity boundary form separate components, making ∂M disconnected.

Definitions for Boolean subtraction and intersection are provided in Section 2 of the supplementary material.

Consistent Faces across Model Variants. When comparing a model M with its modified version $\widehat{M}$, two faces are considered consistent if they have the same edge–vertex topology and the same surface geometry with consistent normals. With a slight abuse of notation, we treat such faces as shared elements in $F_M \cap F_{\widehat{M}}$.

Insight for Accelerating Regeneration. A natural idea for accelerating regeneration is to extract the B-rep "deltas" during generation and reuse them in subsequent regeneration. By applying deltas to the regenerated model, costly geometric operations can be replaced by B-rep face removal and gluing (Fig. 3(c)).

Problem Definition. Unfortunately, in many practical scenarios, reusing these deltas does not behave as expected and can produce results inconsistent with FBR (Fig. 3(d)). In the following sections, we will:

(1) define an appropriate B-rep delta and propose two criteria under which B-rep removal and topological gluing can be safely applied.
(2) outline the proof showing that the glued result is identical to FBR at each step.

4 Locality Criteria and Gluing-based Regeneration

In this section, we define the B-rep deltas recorded during generation and formulate the locality criteria that guarantee their valid reuse during regeneration. For clarity, we restrict the following discussion to a single Type-I STO involving a Boolean union. The notation for all cases is provided in Sections 2 and 4 of the supplementary material.

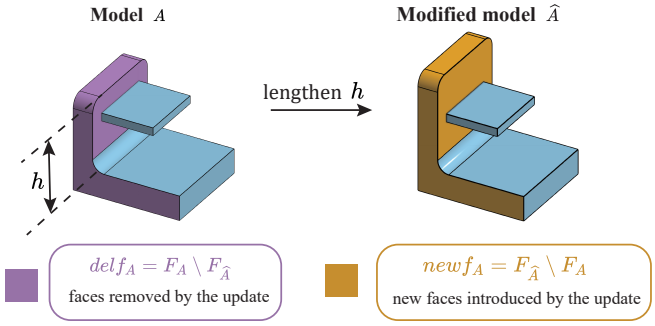


Fig. 6. This figure illustrates the newf and delf regions between a model and its modified version.

4.1 B-rep Delta

Single-Step Notation. For simplicity, we focus on the k^{th} STO and use the following simplified notation:

$$A = M_{k-1}, \quad B = f^{\text{cst}}(M_{k-1}, p_k), \quad C = M_k,$$
$$\widehat{A} = \widehat{M}_{k-1}, \quad \widehat{B} = f^{\text{cst}}(\widehat{M}_{k-1}, \widehat{p}_k), \quad \widehat{C} = \widehat{M}_k.$$

Here, C and $\widehat{C}$ are obtained by applying Boolean union to (A, B) and $(\widehat{A}, \widehat{B})$, respectively.

New Faces under Model Updates. Using the face-consistency convention defined above, shared faces between M and $\widehat{M}$ are identified as $F_M \cap F_{\widehat{M}}$. For $M \in \{A, B\}$, the faces newly appearing in $\widehat{M}$ are defined by

$$\text{newf}_M \triangleq F_{\widehat{M}} \setminus F_M, \quad M \in \{A, B\},$$
$$\text{newf} \triangleq \text{newf}_A \cup \text{newf}_B.$$

The set newf collects the boundary faces that newly appear in the updated models relative to their counterparts in the generation stage. Geometrically, |newf| represents the newly introduced boundary region under the update. Similarly, we define $\text{delf}_M \triangleq F_M \setminus F_{\widehat{M}}$ to collect the faces removed from M by the update. Fig. 6 illustrates newf, delf for a model and its modified version.

B-rep Delta Definition. The transition from (A, B) to C can be viewed as a two-stage boundary modification: the Boolean union first removes the affected faces from F_A and F_B, and then glues in newly generated faces, as illustrated in Fig. 3(b). Formally, we define

$$\text{gen}_A^- \triangleq \{f \in F_A \mid |f| \cap B \neq \emptyset\},$$
$$\text{gen}_B^- \triangleq \{f \in F_B \mid |f| \cap A \neq \emptyset\},$$
$$\text{gen}^+ \triangleq F_C \setminus \big((F_A \setminus \text{gen}_A^-) \cup (F_B \setminus \text{gen}_B^-)\big).$$

Here, set differences involving F_C, F_A, and F_B follow the face correspondence convention defined above. We refer to $\{\text{gen}_A^-, \text{gen}_B^-, \text{gen}^+\}$ as the *B-rep delta* of this modeling step. With this B-rep delta, the output can be recovered by removing gen_A^- and gen_B^- from the input boundaries and topologically gluing in gen^+:

$$\text{Brep}(C) = \text{Glue}\left((F_A \setminus \text{gen}_A^-) \cup (F_B \setminus \text{gen}_B^-) \cup \text{gen}^+\right).$$

The B-rep deltas for Boolean subtraction and intersection require minor adjustments to the definitions above and are provided in Section 2 of the supplementary material.

4.2 Locality Criteria

Scope and Assumptions. We restrict our discussion to valid solid B-reps with manifold boundaries. To keep the main text focused on the generic case, we further impose two simplifying assumptions. Without these assumptions, special cases may arise; their treatment is detailed in Section 2 of the supplementary material.

Assumption 1	Each of $\partial A,\ \partial\widehat{A},\ \partial B,\ \partial\widehat{B}$ is connected.
Assumption 2	$\partial\widehat{A} \cap \partial\widehat{B} \neq \emptyset$.

Under these assumptions, we check whether the stored B-rep delta can be validly reused using the following locality criteria:

Criterion 1	$\mathrm{gen}_A^- \subseteq F_{\widehat{A}}, \quad \mathrm{gen}_B^- \subseteq F_{\widehat{B}}.$
Criterion 2	$\forall f \in \mathrm{newf}_A, \quad \lvert f\rvert \cap \partial\widehat{B} = \emptyset,$
	$\forall f \in \mathrm{newf}_B, \quad \lvert f\rvert \cap \partial\widehat{A} = \emptyset.$

Criterion 1 requires that the faces removed during generation, gen^-, still exist in the updated input solids used for regeneration. Intuitively, this ensures that the original removal region remains available and can be removed again when reusing the stored B-rep delta. Equivalently, for $M \in \{A, B\}$, Criterion 1 can be written as $gen_M^- \cap delf_M = \emptyset$.

Criterion 2 prevents regeneration from introducing new intersection regions relative to the original generation step. Intuitively, this ensures that the newly generated faces gen^+ remain valid during regeneration, thereby avoiding additional intersection computation through FBR.

4.3 Gluing-based Regeneration (GR)

When both criteria are satisfied, we claim that reusing the stored B-rep delta yields exactly the same B-rep as FBR at this step, i.e.,

$$\mathrm{Brep}(\widehat{C}) = \mathrm{Glue}\left((F_{\widehat{A}} \setminus \mathrm{gen}_A^-) \cup (F_{\widehat{B}} \setminus \mathrm{gen}_B^-) \cup \mathrm{gen}^+\right). \tag{1}$$

When the criteria are satisfied, the algorithm avoids BR by applying B-rep removal and topological gluing; we refer to this accelerated step as *gluing-based regeneration* (GR). Otherwise, the algorithm falls back to BR. The resulting stepwise hybrid regeneration algorithm is referred to as *validity-assured regeneration* (VAR).

The applicability of GR is both explicit and broad: even when modifying an STO induces a local topological change, the stored B-rep deltas of later STOs may still be reused. Intuitively, when the modified STO is local and the resulting newf region occupies only a small portion of the model, the regions affected by later STOs during generation are less likely to be influenced by newf, allowing GR to be applied, as illustrated in Fig. 9. In contrast, the method becomes less effective when a feature edit induces a newf region that occupies a large portion of the model.

For clarity, we present the method in the main text at the face level, where B-rep deltas are recorded and reused as face sets. The same idea extends naturally to finer-grained B-rep deltas, such as loop-level deltas, for cases in which the intersection curves remain unchanged during regeneration but the present face-level formulation fails because the intersecting faces undergo loop-level changes and simple face removal and gluing are no longer applicable. Such cases are also illustrated in Fig. 9(e); details are provided in Section 2.3 and Section 3 in the supplementary material.

To establish the validity of our algorithm, we next prove Eq. 1, thereby establishing the correctness of the GR step. Combined with the fallback to BR when the criteria are not satisfied, this guarantees the validity of VAR.

5 Proof Sketch of Validity

We prove that the equality in Eq. 1 holds under the assumptions and criteria stated above, focusing on the Boolean-union case. In terms of point-set topology, this is equivalent to proving the equality of the following point sets:

$$\partial\widehat{C} = \left|\left(F_{\widehat{A}} \setminus \mathrm{gen}_A^-\right) \cup \left(F_{\widehat{B}} \setminus \mathrm{gen}_B^-\right) \cup \mathrm{gen}^+\right|. \tag{2}$$

The extension to arbitrary Boolean operations is provided in Section 3 of the supplementary material.

5.1 Proof Overview

For convenience, let LHS and RHS denote the left-hand side and right-hand side of Eq. 2, respectively. We decompose both sides into the parts associated with $\widehat{A}$ and $\widehat{B}$.

LHS decomposition. The boundary of the regenerated Boolean result can be decomposed as

$$\mathrm{LHS}_A = \partial\widehat{C} \cap \partial\widehat{A},$$
$$\mathrm{LHS}_B = \partial\widehat{C} \cap \partial\widehat{B}.$$

Thus,

$$\mathrm{LHS} = \mathrm{LHS}_A \cup \mathrm{LHS}_B.$$

RHS decomposition. We similarly decompose the reused boundary representation into the parts inherited from A and B. Let

$$\Gamma_A^+ \triangleq |\mathrm{gen}^+| \cap \partial A,$$
$$\Gamma_B^+ \triangleq |\mathrm{gen}^+| \cap \partial B.$$

Then the two parts of the RHS are defined as

$$\mathrm{RHS}_A = \left|F_{\widehat{A}} \setminus \mathrm{gen}_A^-\right| \cup \Gamma_A^+,$$
$$\mathrm{RHS}_B = \left|F_{\widehat{B}} \setminus \mathrm{gen}_B^-\right| \cup \Gamma_B^+.$$

For a Boolean union, the generated boundary gen^+ is inherited from the boundaries of A and B, and hence

$$\mathrm{RHS} = \mathrm{RHS}_A \cup \mathrm{RHS}_B.$$

To establish LHS = RHS, it is enough to prove the two inclusions for each decomposed part:

$$\mathrm{LHS}_i \subseteq \mathrm{RHS}, \qquad \mathrm{RHS}_i \subseteq \mathrm{LHS}, \qquad i \in \{A, B\}.$$

By symmetry between A and B, we only need to verify the A-part:

$$\mathrm{LHS}_A \subseteq \mathrm{RHS}, \qquad \mathrm{RHS}_A \subseteq \mathrm{LHS}.$$

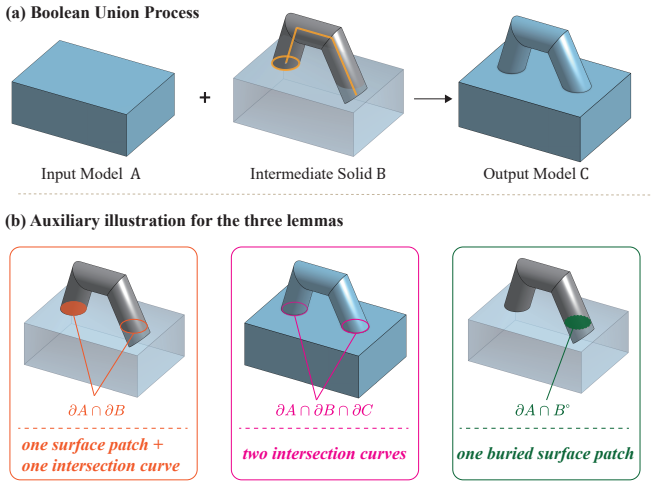


Fig. 7. Illustration of the Boolean union process and the point-set regions used in the three lemmas. (a) Model A is unioned with a swept solid B, generated from a sketch on the top face of A. (b) In this example, $\partial A \cap \partial B$ contains one surface patch and one intersection curve, $\partial A \cap \partial B \cap \partial C$ contains two intersection curves, and $\partial A \cap B^\circ$ contains one buried surface patch, namely a relatively open patch of ∂A lying in the interior of B.

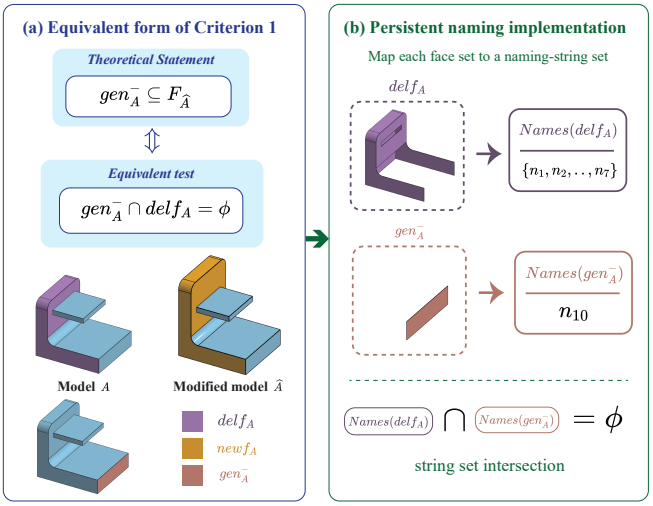


Fig. 8. Criterion 1 reduces to an intersection test between persistent-naming sets, assuming that each face in model A has an unambiguous name. Here, $\{n_1, n_2, \ldots, n_7\}, n_{10}$ denote distinct persistent names; a brief explanation is also provided in Section 6 of the supplementary material.

5.2 Key Lemmas

We first state the key lemmas used in the proof. Their detailed proofs are provided in Section 1.2 of the supplementary material. Fig. 7 shows the point-set regions used in the three lemmas.

Lemma 1. *The intersection between the regenerated input solids remains unchanged:*

$$\partial\widehat{A} \cap \partial\widehat{B} = \partial A \cap \partial B.$$

Intuitively, Criterion 1 *preserves the original boundary-intersection regions, while* Criterion 2 *prevents new boundary-intersection regions from being created during regeneration.*

Lemma 2. *The part of the input–input intersection that remains on the Boolean result is unchanged:*

$$\partial\widehat{A} \cap \partial\widehat{B} \cap \partial\widehat{C} = \partial A \cap \partial B \cap \partial C.$$

Lemma 3. *The boundary region of A contained in the interior of B is preserved during regeneration:*

$$\partial\widehat{A} \cap \widehat{B}^\circ = \partial A \cap B^\circ,$$

where $(\cdot)^\circ$ denotes the interior of a solid.

5.3 Proof of the Main Theorem

We prove the main theorem following the strategy in Sec. 5.1. By symmetry, it suffices to consider the A-part. In the main text, we present the inclusion $\mathrm{LHS}_A \subseteq \mathrm{RHS}$, while the proof of the reverse inclusion $\mathrm{RHS}_A \subseteq \mathrm{LHS}$ is provided separately in Section 1.3 of the supplementary material. Complements $(\cdot)^c$ are taken in $\mathbb{R}^3$.

Proof. Recall that

$$\mathrm{LHS}_A = \partial\widehat{C} \cap \partial\widehat{A}.$$

We first show that

$$\mathrm{LHS}_A \cap \partial A \subseteq \partial C.$$

Decomposing with respect to B°, ∂B, and B^c gives

$$\begin{aligned}
(\mathrm{LHS}_A \cap \partial A) \cap B^\circ &= \partial\widehat{C} \cap \partial\widehat{A} \cap (\partial A \cap B^\circ) \\
&= \partial\widehat{C} \cap \partial\widehat{A} \cap (\partial\widehat{A} \cap \widehat{B}^\circ) = \emptyset. && \text{(Lemma 3)} \\
(\mathrm{LHS}_A \cap \partial A) \cap \partial B &= \partial\widehat{C} \cap \partial\widehat{A} \cap \partial A \cap \partial B \\
&= \partial\widehat{C} \cap \partial\widehat{A} \cap \partial\widehat{B} && \text{(Lemma 1)} \\
&= \partial C \cap \partial A \cap \partial B \subseteq \partial C. && \text{(Lemma 2)} \\
(\mathrm{LHS}_A \cap \partial A) \cap B^c &\subseteq \partial A \cap B^c \subseteq \partial C.
\end{aligned}$$

Therefore,

$$\mathrm{LHS}_A \cap \partial A \subseteq \partial C \;\Rightarrow\; \mathrm{LHS}_A \cap (\partial A \setminus \partial C) = \emptyset.$$

Since $\mathrm{LHS}_A \subseteq \partial\widehat{A}$, this implies

$$\mathrm{LHS}_A \subseteq \partial\widehat{A} \setminus (\partial A \setminus \partial C).$$

By the definition of the B-rep delta,

$$|\mathrm{gen}_A^-| \setminus |\mathrm{gen}^+| \subseteq \partial A \setminus \partial C.$$

Hence,

$$\begin{aligned}
\mathrm{LHS}_A &\subseteq \partial\widehat{A} \setminus (\partial A \setminus \partial C) \\
&\subseteq \partial\widehat{A} \setminus (|\mathrm{gen}_A^-| \setminus |\mathrm{gen}^+|) \\
&\subseteq \left|F_{\widehat{A}} \setminus \mathrm{gen}_A^-\right| \cup |\mathrm{gen}^+| \subseteq \mathrm{RHS}.
\end{aligned}$$

Thus, $\mathrm{LHS}_A \subseteq \mathrm{RHS}$. □

Together with the proof of $\mathrm{RHS}_A \subseteq \mathrm{LHS}$ in the supplementary material and the symmetry between A and B, this establishes LHS = RHS.

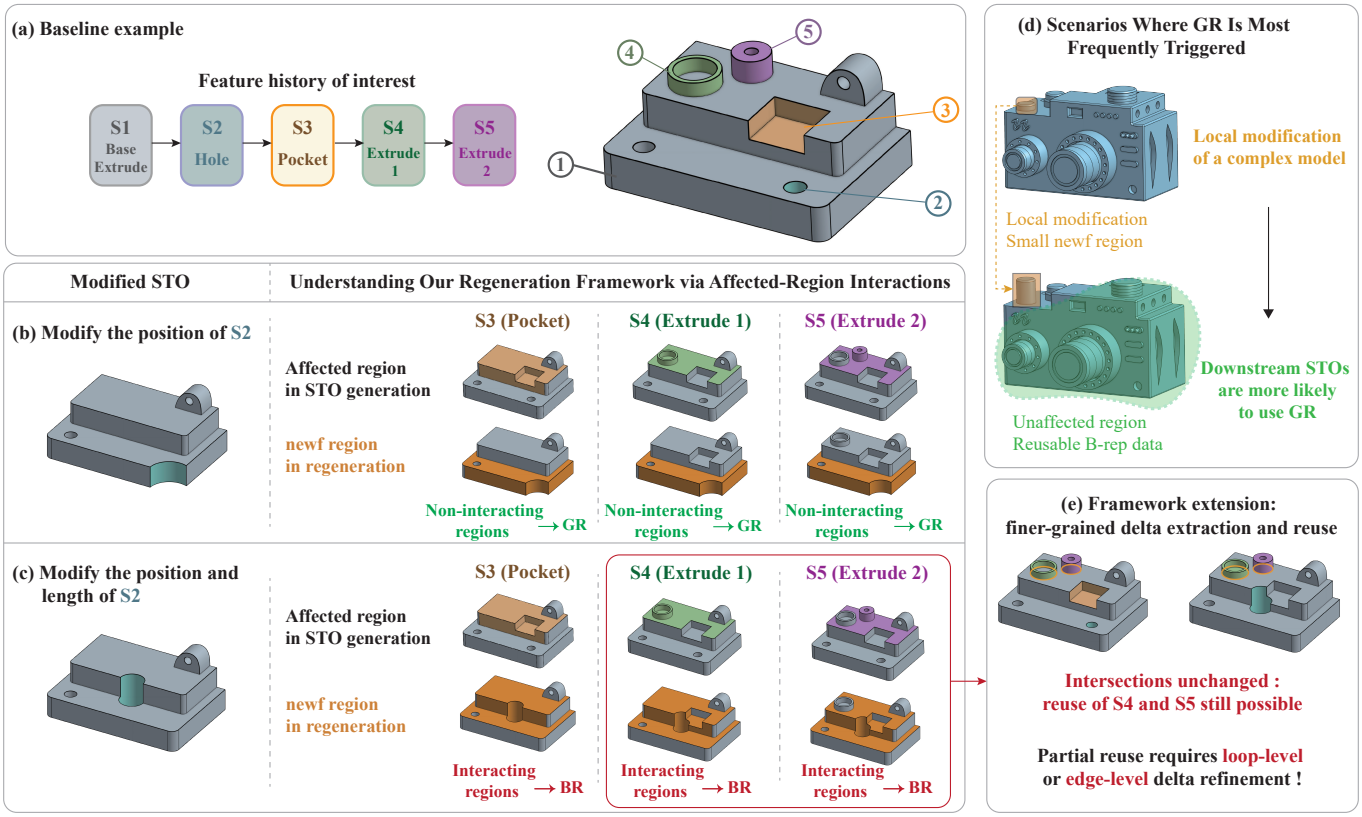


Fig. 9. (a) Key modeling steps of the example used to provide an intuitive understanding of our framework. (b) When the position of the hole in S_2 is modified, GR can still be applied if the resulting newf region does not influence the regions affected by subsequent STOs during generation. Whether such influence exists is determined by our criteria; the illustration here is only intended to provide intuition. (c) In contrast, some modifications do affect the regions to be modified by subsequent STOs, making direct face-level reuse difficult for those later STOs. (d) Intuitively, GR is more likely to apply when the modified feature is local and the resulting newf region remains small, so that STOs outside this region are more likely to reuse their stored B-rep deltas. (e) As shown in (c), S_4 and S_5 cannot directly reuse the faces stored in the generation-time B-rep delta, because the corresponding loops on these faces have changed. We therefore also explore a possible extension based on loop-level delta storage, which may handle certain special yet still easily reusable cases.

6 Validation and Experiment

In this section, we evaluate the validity and efficiency of our regeneration algorithm. All experiments are conducted on a modified version of FreeCAD 0.22.0 dev [FreeCAD contributors 2024] running on Windows.

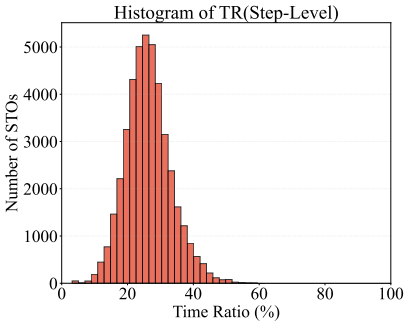
6.1 Experimental Setup

Our Dataset. We build our evaluation dataset from the DeepCAD dataset [Wu et al. 2021], which represents CAD models as CAD construction sequences recorded in JSON files. Each JSON file is converted into a FreeCAD Python script through our conversion pipeline and then executed with FreeCADCmd. We retain only models that contain more than seven STOs, can be reconstructed in FreeCAD, and support local feature modifications. The final evaluation dataset contains 424 valid parametric CAD models.

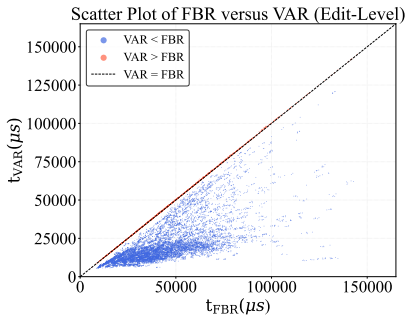
Parameter Modification. To evaluate our method across varying edit magnitudes, we modify every feature except the last one in each model by randomly selecting five scaling factors from the range [0.1, 1.9] and applying them to the feature parameters. This enables a systematic evaluation, under random parameter modifications, of whether the results produced by VAR remain consistent with those of FBR, while also quantifying the runtime difference between VAR and FBR.

Implementation Details. In our experiments, Criterion 1 is implemented through persistent naming of topological faces. Equivalently, for $M \in \{A, B\}$, it reduces to checking whether the naming-string set associated with $\mathrm{gen}_{\bar{M}}$ has empty intersection with that of delf_M, as illustrated in Fig. 8. Criterion 2 is implemented using conservative spatial tests. Although such tests may conservatively enlarge the recomputed region, they do not affect correctness. For topological gluing, we currently use the OpenCascade `BRepBuilderAPI_Sewing` API, mainly to validate correctness and obtain an initial empirical comparison of running time.

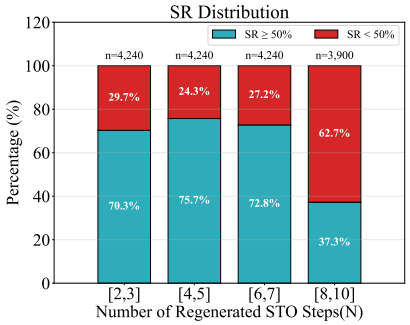
Validity Evaluation. We validate VAR against the FBR baseline at each regenerated STO step by comparing topology counts and paired-entity geometry, with correspondences established using



(a) Histogram of step-level TR, where each STO is regarded as one sample. STOs failing to pass the criteria are excluded. The average is 26.37% and the median is 25.94%.



(b) Scatterplot of edit-level FBR time versus VAR time. Each edit to a model is treated as one data point, with FBR time on the x-axis and VAR time on the y-axis. The percentage of points above the equal line is 25.76%.



(c) SR distribution across different total step (N) groups. Blue segments indicate samples with SR above 50%. Proportions of each segment are labeled inside the bars. Numbers above each bar show the sample size (n) of the group.

Fig. 10. Experimental figures and statistics. Figure (a) shows step-level statistics for successful GR cases. Across 424 models, all modeling steps except the last one in each model are modified five times, yielding 43,019 step-level samples. Figures (b) and (c) show edit-level statistics, with one sample per model modification, giving 16,620 samples in total. For figure (b), the median TR_{global} is 54.32%. Figure (c) shows SR grouped by the total number N of regenerated STOs; for example, the $[4, 5]$ group contains samples requiring 4 or 5 regenerated STOs, of which 24.3% have SR $< 50\%$. This distribution partly reflects feature locality in the dataset. Additional model visualizations and statistics are provided in Section 7 of the supplementary material.

topological identifiers. Across all evaluated cases, VAR matches FBR in both topology and geometry, supporting the validity guarantee in Section 5.

Efficiency Evaluation. For each regenerated STO step j, we record the time for Criterion 1 checking, Criterion 2 checking, topological gluing, and Boolean regeneration (BR), denoted by $t_{\text{c1},j}$, $t_{\text{c2},j}$, $t_{\text{glue},j}$, and $t_{\text{BR},j}$, respectively. The runtime of VAR at step j is defined as

$$t_{\text{VAR},j} = \begin{cases} t_{\text{c1},j} + t_{\text{c2},j} + t_{\text{glue},j}, & \text{if Criteria 1 and 2 pass,} \\ t_{\text{c1},j} + t_{\text{BR},j}, & \text{if Criterion 1 fails,} \\ t_{\text{c1},j} + t_{\text{c2},j} + t_{\text{BR},j}, & \text{otherwise.} \end{cases}$$

We define the step-level time ratio by

$$TR_j = \frac{t_{\text{VAR},j}}{t_{\text{BR},j}} \times 100\%.$$

At the model-edit level, let N_{GR} be the number of regenerated STO steps where GR is applied, and let N be the total number of regenerated STO steps. We define the global saving ratio as

$$SR = \frac{N_{\text{GR}}}{N} \times 100\%.$$

We further define the global time ratio by

$$t_{\text{VAR}} = \sum_j t_{\text{VAR},j}, \quad t_{\text{FBR}} = \sum_j t_{\text{BR},j}, \quad TR_{\text{global}} = \frac{t_{\text{VAR}}}{t_{\text{FBR}}} \times 100\%.$$

6.2 Experimental Results

Step-Level Efficiency of GR. To evaluate the acceleration achieved by GR relative to BR, we collect the step-level time ratio TR_j for every regenerated STO step where the criteria are satisfied and GR is applied, across all parameter modifications of all evaluation models. Steps that fail the criteria are excluded, since they fall back to BR. Fig. 10a shows the distribution of these TR_j values.

Edit-Level Efficiency of VAR. To evaluate the overall regeneration efficiency of VAR, we compare its total runtime with the FBR baseline for each parameter-modification experiment. We compute the overall time ratio TR_{global} for each experiment. Fig. 10b shows the distribution of these edit-level overall time ratios over all parameter modifications of all evaluation models.

Global Saving Ratio. We further evaluate how often VAR replaces BR with GR during regeneration. For each parameter-modification experiment, we compute the global saving ratio SR defined above. Since SR depends only on whether each regenerated STO step is handled by GR or BR, it is independent of the geometric kernel and runtime performance. Fig. 10c shows the distribution of SR over all parameter modifications of all evaluation models.

6.3 Summary of Results

The results show that, for successful GR steps, the average time ratio is 26.37%, indicating a substantial speedup over BR. Moreover, when the modification is confined to a local feature with relatively small parameter changes, the saving ratio can remain high, allowing VAR to achieve greater time savings than FBR. Since DeepCAD models have relatively short histories and are often geometrically simple, modifying certain features may induce nonlocal changes that affect the global geometry. As a result, 25.76% of the modification cases have $TR_{\text{global}} > 1.0$, though the excess over 1.0 is typically small. The reason is that our criteria are still evaluated for nearly all regenerated steps, but GR is rarely triggered, so the model eventually falls back to BR with extra checking overhead. To further complement our experimental evaluation, we additionally test two models with longer histories, containing 37 and 53 STO steps, respectively. Detailed results are provided in Section 5 of the supplementary material.

7 Conclusion

In this paper, we present a validity-assured regeneration algorithm that captures B-rep delta information during CAD model generation and reuses it to accelerate regeneration. When the two proposed locality criteria are satisfied, these deltas can be reused through GR in subsequent regenerations, thereby improving efficiency. We believe this work helps establish regeneration as a more systematic research problem, highlights its practical importance, and provides an implementable foundation for further study in this area.

The current framework is limited by its reliance on B-rep delta reuse at the face or loop level, and is therefore less effective for edits that induce large-scale geometric and topological changes across the model. Future work will investigate finer-grained reuse, such as edge-level delta representations, and explore how the proposed ideas can be integrated into hybrid workflows that combine feature-based and direct modeling.

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